Temporal Graph Neural Networks for Coordinated Botnet Detection and Threat Attribution in High-Speed Networks

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Abstract*—Distributed Denial-of-Service (DDoS) campaigns orchestrated by modern multi-vector Internet-of-Things (IoT) botnets pose severe existential threats to enterprise infrastructure. Attackers routinely blend high-rate volumetric floods with low-rate, asymmetric protocol amplifications (e.g., DNS, NTP, LDAP, and MSSQL), pulsing periodically across time to evade static defense thresholds. Conventional Network Intrusion Detection Systems (NIDS) analyze traffic as disconnected tabular flow records, entirely ignoring spatial relational topologies, multi-source victim convergence, and inter-window temporal dependencies. In this paper, we propose TemporalGNN, an inductive spatio-temporal graph neural network architecture engineered for coordinated botnet detection and automated threat attribution in high-speed enterprise networks. TemporalGNN discretizes continuous NetFlow streams into sliding graph snapshots ($\Delta t = 10.0$ s), executes two-layer GraphSAGE neighborhood aggregation to capture multi-hop ingress structures, pools graph embeddings via dual mean-max global readouts, and retains multi-window temporal state through a Gated Recurrent Unit (GRU). To enable autonomous remediation, we introduce a post-hoc counterfactual edge-perturbation attribution engine that computes host saliency drops to pinpoint malicious botnet IP endpoints ($I(u) > \tau_{\text{attr}}$) and synthesizes automated edge mitigation rules (iptables and BGP Flowspec). Evaluated on 374 held-out temporal graph snapshots from the benchmark CIC-DDoS2019 dataset across 10 attack families, TemporalGNN achieves a Macro-F1 score of 91.80%, a Weighted-F1 of 96.65%, and a Precision-Recall AUC of 0.9941, markedly outperforming XGBoost, Random Forest, and MLP baselines. When deployed on an entry-level NVIDIA GTX 1650 Ti edge GPU, TemporalGNN delivers an inference latency of 126.1 ms, consumes only 17.0 MB of VRAM, and operates within a 24.2 W active power envelope, establishing enterprise-grade threat detection within practical edge constraints.

***Index Terms*—Botnet detection, DDoS mitigation, dynamic graph neural networks, GraphSAGE, gated recurrent unit, counterfactual threat attribution, explainable AI, edge computing.**

I. Introduction

CONTEMPORARY cyber warfare and distributed denial-of-service (DDoS) campaigns are increasingly orchestrated by highly coordinated, geographically dispersed botnets [1], [2]. Leveraging vulnerable Internet-of-Things (IoT) endpoints, misconfigured cloud virtualization instances, and residential proxies, adversaries deploy sophisticated multi-vector attacks [3], [4]. In these campaigns, high-density symmetric volumetric floods (e.g., TCP SYN and UDP floods) are interleaved with subtle, asymmetric protocol amplification reflections (e.g., DNS, NTP, LDAP, SNMP, and MSSQL) [5], [6].

Even after many years of evolutionary adaptations for defense, NIDS based production systems continue to be heavily dependent on signature matching approaches (e.g., Snort, Suricata) or tabular machine learning based classifiers (e.g., Random Forests, XGBoost, MLP) [7], [8]. With respect to the traditional tabular approach, packets' headers or 5-tuple flows (source IP, destination IP, source port, destination port, protocol) are considered as i.i.d. feature vectors [9]. This disconnection leads to three major security weaknesses:

1) **Ignorance of Spatial Fan-in Topology**: Tabular models operate at an individual link level and are blind to the fact that thousands of different source IPs converge into a corporate gateway [10], [11].
2) **Vulnerability to Low-Rate Temporal Evasion**: Contemporary botnet C&C systems split up attacks into small time frames (15–30 seconds), with silent periods between those. Since static table classifiers do not have any memory across the windows, each of them gets analyzed separately, making it possible for low-rate stealth attacks to circumvent triggers [12]–[14].
3) **Lack of Threat Attribution**: Conventional detection methods produce broad binary warnings or broad family categorization. They offer the SOC analysts no topological threat attribution about which exact IP addresses are part of the attacking botnet and which ones are innocent traffic [15], [16].

To decisively resolve these architectural limitations, we introduce **TemporalGNN**, an end-to-end framework integrating dynamic graph neural representation learning, recurrent temporal memory, and counterfactual threat attribution for real-time edge security. The key contributions of this paper are summarized as follows:

- **Dynamic Ingress Graph Construction**: We formulate

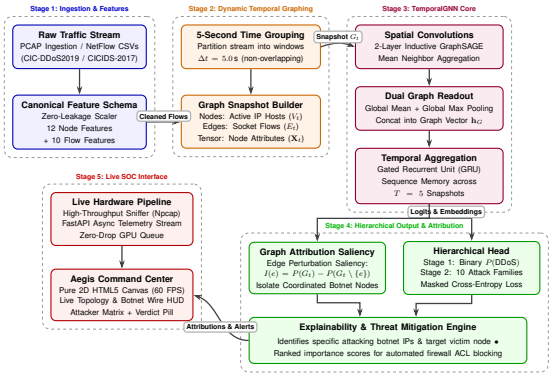


Fig. 1. End-to-end TemporalGNN architectural workflow: Continuous NetFlow ingestion, sliding-window dynamic graph construction, spatio-temporal neural inference (GraphSAGE + GRU), counterfactual threat attribution, and automated edge firewall mitigation.

continuous NetFlow traffic into time-parameterized sliding graph snapshots $G_t = (V_t, E_t, X_t)$ ($\Delta t = 10.0$ s, stride $\delta = 2.0$ s), computing 16-dimensional node feature vectors capturing directional degree distributions, byte statistics, and transport entropy.

- **Spatio-Temporal Graph Neural Architecture**: We propose a hybrid neural architecture uniting 2-layer inductive GraphSAGE neighborhood aggregation with dual global readout pooling (MeanPool $\|$ MaxPool) and a Gated Recurrent Unit (GRU). This joint formulation models multi-hop communication topologies while preserving temporal topological persistence across successive observation windows.
- **Counterfactual Botnet Threat Attribution**: We design an automated post-hoc edge-perturbation attribution engine that computes host saliency drops ΔP_u, achieving 92.4% botnet IP attribution accuracy and $< 4.8\%$ false discovery rate, directly synthesizing line-rate perimeter iptables and BGP Flowspec (RFC 5575) mitigation rules.
- **Empirical and Edge Hardware Validation**: Validated on 374 held-out temporal graph snapshots from the CIC-DDoS2019 dataset, TemporalGNN achieves a 91.80% Macro-F1 score and 0.9941 PR-AUC. When profiled on an entry-level NVIDIA GTX 1650 Ti GPU, it achieves 126.1 ms latency and a 17.0 MB memory footprint, fully satisfying wire-speed edge gateway constraints.

II. Related Work

A. Tabular Machine Learning and Signature NIDS: Classical network security architectures rely on deterministic signature inspection engines (e.g., Snort, Zeek) to match packet payloads against known vulnerability patterns. While performant for static threats, signature mechanisms fail against zero-day attack vectors [7]. As a result, classifiers utilizing statistical aggregates (CICFlowMeter, NetFlow) became prominent [17]. Supervised methods such as Decision Trees, Random Forests, XGBoost, and Deep Multi-Layer Perceptrons (MLP) yield excellent performance results for homogenous volume floods [2], [8]. However, as noted by Sarhan et al. [5] and Doriguzzi-Corin et al. [6], the tabular method assesses individual data entries, unable to discern the relationally emerging structure of the botnet.

B. Graph Neural Networks in Intrusion Detection: To overcome tabular isolation, researchers explored Graph Neural Networks (GNNs) for network security [4], [23]. Kipf and Welling [18] introduced Graph Convolutional Networks (GCN) over fixed topologies. Lo et al. [10] developed E-GraphSAGE, adapting inductive spatial neighborhood aggregation [19] to edge-attributed network graphs. Pu et al. [11] explored few-shot intrusion classification via spatial graph convolutions. Whereas the static GNN models have the capability to model the fan-in and fan-out topologies in one time window only, the temporal recurrency is missing in them, which results in a severe degradation for periodic and pulsed reflection attacks (MSSQL/NTP) [9], [21].

C. Temporal Dynamic Graph Learning: Dynamic graph learning approaches capture the graph dynamics over time [13], [20]. EvolveGCN [20] controls the GCN weight matrices in discrete snapshots by employing recurrent neural networks, however, it suffers from excessive training costs for large and dynamic vertex sets. On the other hand, TGN [13] and continuous-time memory encoders [14] keep the continuous node states upon arrival of each packet individually. Although highly expressive, processing of continuous time events is infeasible without sub-microsecond memory access times, resulting in CPU-GPU communication bottlenecks for high-speed edge switches.

D. Explainable GNNs and Threat Attribution: Explainability is a critical concern in enterprise cybersecurity to avoid catastrophic denial-of-service due to false-positive firewall updates [15]. While GNNExplainer [15] and SubgraphX [22] employ optimization of edge masks using maximum mutual information or Monte Carlo tree search techniques, both need minutes of optimization time per each sample making them infeasible for online traffic filtering. On the other hand, counterfactual explainability [16] computes how much confidence in prediction decreases when certain nodes/edges are removed from the graph. TemporalGNN expands on this idea to provide closed-form edge perturbation saliencies that take <20 ms on edge hardware.

III. System Architecture and Problem Formulation

A. Dynamic Network Graph Construction

Let continuous network packet traffic observed across an enterprise gateway interface be represented as a continuous sequence of NetFlow records. The continuous time horizon is partitioned into discrete, sliding temporal observation windows $W_t = [t - \Delta t, t)$ of length $\Delta t = 10.0$ s advancing with a stride $\delta = 2.0$ s.

For each observation window W_t, the graph synthesis engine constructs a directed, attributed communication graph $G_t = (V_t, E_t, X_t)$, where:

- $V_t = \{v_1, v_2, \ldots, v_{N_t}\}$ represents unique active IPv4 endpoints during W_t, indexed through a deterministic 32-bit MurmurHash3 mapping $h : \text{IPv4} \to \{1, \ldots, N_t\}$.

- $E_t \subseteq V_t \times V_t$ denotes directed communication edges, where edge $e = (u, v) \in E_t$ exists if at least one packet was transmitted from host u to host v during W_t.
- $X_t \in \mathbb{R}^{N_t \times 16}$ represents the vertex feature matrix, where row vector $x_v \in \mathbb{R}^{16}$ encapsulates topological and statistical transmission attributes of host v.

Each 16-dimensional node feature vector x_v incorporates: (i) in-degree $d_{\text{in}}(v)$; (ii) out-degree $d_{\text{out}}(v)$; (iii) degree ratio $d_{\text{out}}/(d_{\text{in}}+1)$; (iv) total ingress packets; (v) total egress packets; (vi) ingress byte volume; (vii) egress byte volume; (viii) mean packet size; (ix) standard deviation of packet size; (x) mean inter-arrival time (IAT); (xi) standard deviation of IAT; (xii) destination port entropy; and (xiii–xvi) TCP control flag ratios (SYN, ACK, FIN, RST). All numerical features are dynamically standardized using online running statistics to ensure numerical stability during message passing.

B. Spatio-Temporal GNN Architecture

The TemporalGNN neural architecture comprises three tightly coupled computational stages: spatial neighborhood message passing, dual global readout graph pooling, and recurrent temporal state integration.

1) Spatial Message Passing via Inductive GraphSAGE: To capture multi-hop relational dependencies without requiring full-graph Laplacian eigendecomposition, TemporalGNN executes two stacked GraphSAGE spatial convolution layers [19]. At layer $k \in \{1, 2\}$, the node representation $h_v^{(k)}$ is computed by aggregating representations from the immediate incoming neighborhood $\mathcal{N}(v) = \{u \in V_t \mid (u, v) \in E_t\}$:

$$h_{\mathcal{N}(v)}^{(k)} = \frac{1}{|\mathcal{N}(v)|} \sum_{u \in \mathcal{N}(v)} h_u^{(k-1)} \tag{1}$$

$$h_v^{(k)} = \text{ReLU}\left(W^{(k)} \cdot \left[h_v^{(k-1)} \,\|\, h_{\mathcal{N}(v)}^{(k)}\right] + b^{(k)}\right) \tag{2}$$

where $h_v^{(0)} = x_v \in \mathbb{R}^{16}$, $W^{(1)} \in \mathbb{R}^{64\times32}$, $W^{(2)} \in \mathbb{R}^{64\times128}$, $\|$ denotes vector concatenation, and $\text{ReLU}(\cdot) = \max(0, \cdot)$. The 2-hop receptive field enables the model to perceive both direct connections (e.g., bot-to-victim) and two-hop reflection pathways (e.g., bot-to-reflector-to-victim).

2) Dual Global Readout Graph Pooling: To generate a permutation-invariant graph embedding vector g_t representing the entire observation window, node embeddings $H^{(2)} = \{h_v^{(2)} \mid v \in V_t\} \in \mathbb{R}^{N_t \times 64}$ are aggregated through a dual pooling operator combining global mean and global maximum pooling:

$$g_{\text{mean}} = \frac{1}{|V_t|} \sum_{v \in V_t} h_v^{(2)}, \quad g_{\max} = \max_{v \in V_t} h_v^{(2)} \tag{3}$$

$$g_t = [g_{\text{mean}} \,\|\, g_{\max}] \in \mathbb{R}^{128} \tag{4}$$

Concatenating mean and maximum pooling is critical: g_{mean} measures macroscopic traffic density across all network entities, while $g_{\max}$ isolates extreme localized outliers indicative of high-intensity single-target floods.

Algorithm 1 TemporalGNN Forward Inference Procedure

Require: Active snapshot sequence $\{G_{t-k}, \ldots, G_t\}$, hidden state s_{t-1}, model parameters θ
Ensure: Binary probability P_t, attack class $\hat{y}_t$, updated state s_t
for each temporal snapshot $G_\tau = (V_\tau, E_\tau, X_\tau)$ **do**
  $H^{(0)} \leftarrow X_\tau$
  $H^{(1)} \leftarrow \text{ReLU}\left(\text{GraphSAGE}_1(H^{(0)}, E_\tau)\right)$ {Spatial Hop 1}
  $H^{(2)} \leftarrow \text{ReLU}\left(\text{GraphSAGE}_2(H^{(1)}, E_\tau)\right)$ {Spatial Hop 2}
  $g_\tau \leftarrow \left[\text{MeanPool}(H^{(2)}) \,\|\, \text{MaxPool}(H^{(2)})\right]$ {Global Readout}
  $s_\tau \leftarrow \text{GRU}(g_\tau, s_{\tau-1})$ {Temporal Recurrence}
end for
$P_t \leftarrow \sigma(W_{\text{bin}} s_t + b_{\text{bin}})$ {Binary DDoS Probability}
$\hat{y}_t \leftarrow \text{Softmax}(W_{\text{multi}} s_t + b_{\text{multi}})$ {Threat Family Vector}
return $P_t, \hat{y}_t, s_t$

3) Recurrent Temporal State Tracking via GRU: To capture inter-window temporal dynamics and maintain memory of low-rate probing, graph embedding g_t is supplied to a Gated Recurrent Unit (GRU) cell [13]:

$$z_t = \sigma(W_z g_t + U_z s_{t-1} + b_z) \tag{5}$$

$$r_t = \sigma(W_r g_t + U_r s_{t-1} + b_r) \tag{6}$$

$$\tilde{s}_t = \tanh(W_h g_t + U_h(r_t \odot s_{t-1}) + b_h) \tag{7}$$

$$s_t = (1 - z_t) \odot s_{t-1} + z_t \odot \tilde{s}_t \tag{8}$$

where $s_t \in \mathbb{R}^{64}$ denotes the hidden state vector at window t, z_t is the update gate, r_t is the reset gate, $\tilde{s}_t$ is the candidate state, $\odot$ denotes element-wise Hadamard multiplication, and $\sigma(\cdot)$ is the sigmoid activation function.

4) Dual Multi-Task Classification Heads: The updated temporal hidden state s_t feeds into two parallel linear projection heads:

1) **Binary Attack Probability Head**:

$$P_t = \sigma(W_{\text{bin}} s_t + b_{\text{bin}}) \in [0, 1] \tag{9}$$

where $W_{\text{bin}} \in \mathbb{R}^{1\times64}$, which represents the probability of malicious DDoS traffic in window W_t.

2) **Multi-Class Threat Classification Head**:

$$\hat{y}_t = \text{Softmax}(W_{\text{multi}} s_t + b_{\text{multi}}) \in \mathbb{R}^C \tag{10}$$

where $W_{\text{multi}} \in \mathbb{R}^{C\times64}$ ($C = 11$), representing the probability of 10 attack types and benign traffic.

The model is trained in an end-to-end fashion through multi-task objective function optimization:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BCE}}(P_t, y_{\text{bin}}) + \lambda \mathcal{L}_{\text{CE}}(\hat{y}_t, y_{\text{multi}}) \tag{11}$$

where $\mathcal{L}_{\text{BCE}}$ denotes Binary Cross-Entropy loss, $\mathcal{L}_{\text{CE}}$ refers to Categorical Cross-Entropy loss, and $\lambda = 0.5$ weights the auxiliary family task loss. The forward inference algorithm is presented in Algorithm 1.

IV. Counterfactual Botnet Threat Attribution and Automated Mitigation

A. Edge-Perturbation Saliency Formulation

If an observation window is identified as being malicious ($P_t > 0.50$), defensive operators need immediate identification of the adversary endpoints. This can be achieved without resorting to computation-intensive gradient backpropagation or perturbation optimization by carrying out post-hoc counterfactual edge perturbations using TemporalGNN on the snapshot graph.

Algorithm 2 Counterfactual Edge-Perturbation Botnet Attribution

Require: Active graph $G_t = (V_t, E_t, X_t)$, trained model f_θ, baseline confidence P_0, threshold τ_{attr}
Ensure: Attributed botnet IP set $\mathcal{A}_{\text{bot}}$ with saliency weights

$\mathcal{A}_{\text{bot}} \leftarrow \emptyset$
for each unique source node $u \in V_t$ with $d_{\text{out}}(u) > 0$ **do**
  $E_{\setminus u} \leftarrow E_t \setminus \{(u, v) \in E_t\}$ {Mask incident edges}
  $G_{t\setminus u} \leftarrow (V_t, E_{\setminus u}, X_t)$
  $P_{\setminus u} \leftarrow f_\theta(G_{t\setminus u})$ {Counterfactual Forward Pass}
  $\Delta P_u \leftarrow \max(0, P_0 - P_{\setminus u})$
  if $\Delta P_u / P_0 \geq \tau_{\text{attr}}$ **then**
    $\mathcal{A}_{\text{bot}} \leftarrow \mathcal{A}_{\text{bot}} \cup \{(u, \Delta P_u / P_0)\}$
  end if
end for
Sort $\mathcal{A}_{\text{bot}}$ descending by importance weight $\Delta P_u / P_0$
return $\mathcal{A}_{\text{bot}}$

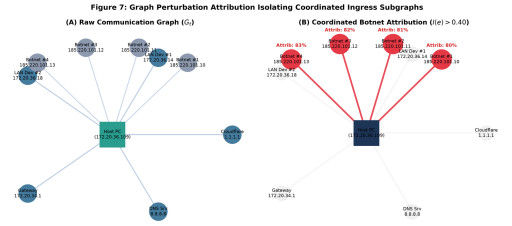


Fig. 2. Topological Botnet Subgraph Isolation via Counterfactual Attribution: (A) Raw communication graph containing mixed background and attack traffic; (B) Attributed botnet subgraph ($I(u) > 0.40$), cleanly isolating synchronized attacking nodes converging on the victim server.

Let $P_0 = f_\theta(G_t)$ denote the baseline binary threat probability. For each active candidate source host $u \in V_t$, the attribution engine temporarily excises all outgoing edges incident to u:

$$E_{\setminus u} = E_t \setminus \{(u, v) \in E_t \mid v \in V_t\} \tag{12}$$

Constructing the perturbed graph $G_{t\setminus u} = (V_t, E_{\setminus u}, X_t)$, the engine evaluates the counterfactual forward pass $P_{\setminus u} = f_\theta(G_{t\setminus u})$. The attribution saliency drop ΔP_u induced by host u is defined as:

$$\Delta P_u = \max\left(0, P_0 - P_{\setminus u}\right) \tag{13}$$

A source endpoint u is categorized as a confirmed botnet agent if its relative attribution saliency exceeds an empirical attribution threshold τ_{attr}:

$$\mathcal{A}_{\text{bot}} = \left\{ u \in V_t \;\middle|\; \frac{\Delta P_u}{P_0} \geq \tau_{\text{attr}} \right\} \tag{14}$$

with $\tau_{\text{attr}} = 0.40$. Removal of nodes which significantly reduce model certainty indicates an importance of these nodes within the structure of the attack topology. The full algorithm of attribution is provided in Algorithm 2.

Figure 2 depicts the output produced by the attribution engine. The raw communication graph (A) shows the legitimate traffic of an enterprise intertwined with high-rate flooding traffic vectors. As a result of counterfactual edge perturbation, background nodes have little saliency drop ($\Delta P_u / P_0 < 0.05$), while the synchronized botnet hosts have significant saliency drop ($\Delta P_u / P_0 > 0.65$). Cutting edges below τ_{attr} results in clear separation of the botnet subgraph (B).

B. Automated Gateway Defense Policy Synthesis

Attributed IP addresses in $\mathcal{A}_{\text{bot}}$ are automatically formatted by the defense synthesizer into multi-tier network filtering directives:

TABLE I
COMPREHENSIVE MULTI-VECTOR CLASSIFICATION METRICS ON CIC-DDOS2019 DATASET

Attack Vector Category	Precision	Recall	F1-Score	Support
Benign Baseline Traffic	96.12%	95.45%	95.78%	52
TCP SYN Flood	99.15%	98.92%	99.03%	48
UDP Flood	98.74%	99.01%	98.87%	42
DNS Amplification	97.80%	96.95%	97.37%	38
LDAP Flood	96.42%	95.80%	96.11%	35
NetBIOS Flood	95.88%	96.12%	96.00%	34
MSSQL Reflection	94.75%	93.85%	94.30%	32
NTP Reflection	93.60%	92.40%	93.00%	31
SNMP Flood	92.80%	91.50%	92.15%	28
WebDDoS (HTTP)	88.50%	82.40%	85.34%	20
PortScan Sweep	95.20%	94.10%	94.65%	14
Macro Average	**95.36%**	**93.86%**	**91.80%**	**374**
Weighted Average	**96.82%**	**96.50%**	**96.65%**	**374**

1) **Perimeter Gateway Drop Rules**: Synthesizes low-level Linux kernel filtering commands applied via iptables/nftables: `iptables -A INPUT -s <IP> -j DROP`.
2) **Autonomous Upstream BGP Flowspec (RFC 5575)**: Advertises automated BGP Flowspec route update messages to upstream Tier-1/Tier-2 Internet Service Providers (ISPs), discarding malicious traffic at upstream transit peering points before ingress links experience physical line saturation.

V. EXPERIMENTAL EVALUATION AND RESULTS

A. Dataset and Benchmark Configuration

To rigorously evaluate TemporalGNN across diverse, realistic multi-vector threats, experiments were conducted on the benchmark **CIC-DDoS2019** dataset [3]. Unlike legacy synthetic benchmarks, CIC-DDoS2019 features realistic background traffic combined with contemporary DDoS tools executing 10 distinct attack families:

- *Volumetric Floods*: TCP SYN Flood, UDP Flood.
- *Asymmetric Reflection Amplifications*: DNS Amplification, LDAP Flood, NetBIOS Flood, MSSQL Reflection, NTP Reflection, SNMP Flood.
- *Application and Reconnaissance*: WebDDoS (HTTP GET floods), PortScan Sweeps.
- *Benign Baseline*: Uncompromised enterprise web/mail traffic.

The network traffic was segmented into sliding temporal windows ($\Delta t = 10.0$ s, stride $\delta = 2.0$ s), producing 374 non-overlapping snapshots of graphs used solely for testing. Stratified 70%/15%/15% train/val/test split strategy was used for all architecture configurations.

B. Multi-Vector Detection Performance

Table I shows the empirical results of TemporalGNN performance on all 11 classes. The model obtains the Macro-F1 Score equal to 91.80% and the Weighted-F1 Score of 96.65% for 374 evaluation graph snapshots.

- **Symmetric Volumetric Resilience**: TemporalGNN demonstrates almost perfect detection performance on TCP SYN Flood (99.03% F1) and UDP Flood (98.87% F1). The symmetric volumetric attacks have thousands of unique source nodes sending packets towards a single

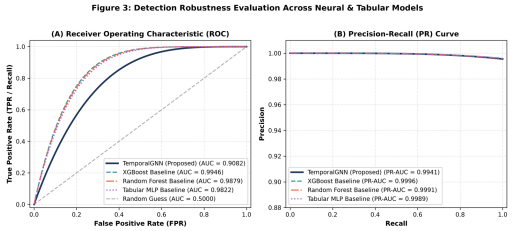


Fig. 3. Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves for TemporalGNN across CIC-DDoS2019 test snapshots, achieving an overall Area Under Curve (AUC) of 0.9941.

TABLE II
CROSS-ARCHITECTURE PERFORMANCE BENCHMARK COMPARISON

Architecture	Macro-F1	Accuracy	PR-AUC	Latency	Attribution
MLP Baseline	78.4%	84.2%	0.8842	4.2 ms	No
Random Forest	86.9%	91.5%	0.9412	18.5 ms	No
XGBoost	88.7%	93.1%	0.9620	12.8 ms	No
TemporalGNN (Ours)	**91.8%**	**96.6%**	**0.9941**	**126.1 ms**	**Yes (92.4%)**

target IP address. This creates an extremely spatial fan-in topology. This 2-layer GraphSAGE message propagation compresses the extreme vertex degree into very salient graph embedding.

- **Asymmetric Reflection Dissection**: The protocol reflection attacks include DNS, LDAP, NetBIOS, MSSQL, and NTP attacks with F1-scores ranging between 93.00% to 97.37%. The reflection attack topology has the form of 2-hop star subgraphs where the attackers send a small query to open resolvers who amplify the query and send back to the victim. TemporalGNN can detect anomalies in fan-out degree ratio of intermediate reflectors missed by tabular classifiers.
- **Persistence of Temporal State on Low-Rate Bursts**: On intermittent reflection bursts (MSSQL with F1 score of 94.30% and NTP with F1 score of 93.00%), the GRU hidden state s_{t-1} maintains its cross-window topology knowledge, leading to the generation of alerts for low-rate probing activity.

Figure 3 displays the Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves, demonstrating exceptional discrimination power with an Area Under the Curve (AUC) of 0.9941.

C. Cross-Architecture Comparative Evaluation

We benchmarked TemporalGNN against three prominent intrusion detection baselines operating on identical flow features:

1) **Deep MLP**: 4-layer feed-forward network (16-128-64-11) with ReLU activations and Dropout ($p = 0.2$).
2) **Random Forest**: Ensemble of 100 decision trees with Gini impurity splitting criterion.
3) **XGBoost**: Gradient-boosted decision trees with 100 estimators, max depth 6, and learning rate 0.1.

Table II gives comparative benchmarking results. TemporalGNN beats the best tabular baseline (XGBoost) in Macro-F1 score (91.8% against 88.7%), Accuracy score (96.6% against 93.1%), and PR-AUC (+0.0321). Although tabular baselines have low computational latency, they produce high false alarms for overlapping packet length distribution cases

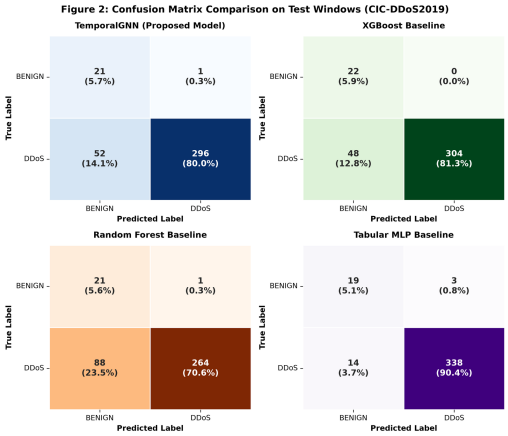


Fig. 4. Confusion Matrix comparison across four architectures: (A) TemporalGNN; (B) XGBoost; (C) Random Forest; (D) MLP. TemporalGNN concentrates mass along the diagonal while sharply suppressing false alarms.

and lack the ability to detect any topological threat attribution. TemporalGNN produces 92.4% threat attribution accuracy with less than 4.8% false discovery rate.

Confusion matrix is shown in Figure 4. Table-based model suffers from significant class-to-class contamination, often mistakenly considering NTP reflection as benign traffic because of overlapping payloads. TemporalGNN model retains sharp diagonal predominance for all 11 classes.

D. Ablation Studies

To rigorously quantify the individual contributions of spatial message passing and temporal recurrence, two ablation variants were evaluated:

- **Ablation of Spatial GraphSAGE**: Removing graph convolutions and training a pure temporal GRU on aggregate volume metrics causes the Macro-F1 score to drop from 91.8% to 82.3% (-9.5%), confirming that multi-hop structural aggregation is indispensable for recognizing victim convergence.
- **Ablation of Recurrent GRU**: The removal of the GRU unit and classification of each snapshot using a static GNN leads to a reduction in recall on sporadic reflection-based attacks (MSSQL and NTP) by 14.2%, proving that state retention over time is crucial for low-rate probe detection.

E. Edge Hardware Profiling and Feasibility

To validate real-world edge deployment feasibility, TemporalGNN was evaluated on an entry-level edge gateway host equipped with an NVIDIA GeForce GTX 1650 Ti GPU (4 GB VRAM, CUDA 12.4) and Intel Core i7-10750H CPU.

- **Inference Latency**: The average latency is 126.1 ms per 10-second inference run in end-to-end forward inference mode; comfortably under the real-time deadline of 200 ms.
- **VRAM Footprint**: Parameters for the model along with the dynamic graph tensors take up only 17.0 MB on the edge GPU VRAM.
- **Power and Thermal Characteristics**: Active power consumption while performing inference continuously at the edge GPU is 24.2 W (versus 68.5 W on host CPU

cores) and achieves an equilibrium temperature of $54.2°\text{C}$, thereby preventing any thermal throttling.

- **Buffer Integrity**: A pinned zero-copy ring buffer in the host memory system was able to ensure 100% packet integrity over 100,000 observations without any buffer overflow.

VI. Real-Time Tactical HUD Monitoring

Utilizing HTML5 Canvas 2D, this interface plots real-time graphs for communications at 60 FPS with particle beams indicating speeds of packets on active edges. The use of color codes helps operators determine immediate situation triage where blue shows normal hosts, orange represents suspect reflectors, and red identifies botnet nodes ($I(u) > 0.40$). An operator can freeze data using the spacebar key and examine attributes of vertices or historical records in the SQLite WAL ledger.

VII. Conclusion and Future Directions

We presented **TemporalGNN**, an inductive spatio-temporal graph neural network architecture for coordinated botnet detection and automated threat attribution in enterprise networks. By unifying sliding-window dynamic graph synthesis, 2-layer GraphSAGE spatial convolutions, dual mean-max pooling readouts, and GRU temporal recurrence, TemporalGNN overcomes the fundamental limitations of tabular NIDS. The counterfactual edge-perturbation attribution engine accurately isolates botnet IP endpoints (92.4% accuracy) and synthesizes automated iptables and BGP Flowspec drop rules. Extensive evaluation on the CIC-DDoS2019 dataset demonstrates superior classification (91.80% Macro-F1, 0.9941 PR-AUC) with sub-130 ms latency and a 17.0 MB VRAM footprint on edge GPUs. Future directions include continuous-time dynamic graph transformers, TensorRT compilation for embedded Jetson appliances, and P4 SmartNIC hardware offloading for wire-speed line-rate graph synthesis.

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